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AI-Driven Synergistic Optimization of CO_2-Associated Gas Injection and CCUS in Tight Oil Reservoirs

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Abstract

Tight reservoirs, characterized by extremely low porosity and permeability as well as restricted fluid mobility, have long faced challenges such as low recovery and high development difficulty. CO_2-mixed associated gas drive has a broad application prospect, which acts in the function of an efficient development technology with both enhanced recovery and carbon sequestration capacity. The regulation of gas injection parameters involves the synergistic effect of multiple variables, but the traditional optimization methods have the problems of insufficient adaptability in the regulation of gas injection parameters and scheme design, which is difficult to meet the actual needs. This paper introduces artificial neural network and genetic algorithm to develop an intelligent optimization model, conducting research on residual distribution prediction, identification of controlling factors, optimization of gas injection timing, and the synergistic optimization of recovery and sequestration efficiency. The findings demonstrate that the residual oil in the target block is categorized into five types: imperfect residual oil in the well network, residual oil in the thin differential layer, residual oil masked by closed faults, residual oil not controlled by the well network, and residual oil controlled by sporadic sands. The interaction of gas injection rate and bottomhole flow pressure has a significant effect on both recovery and sequestration ratio. Among the factors, bottomhole flow pressure is the most important factor affecting the recovery, and gas injection rate is the crucial factor affecting the sequestration ratio. Reasonable timing of gas injection is the crucial factor in improving recovery. When the pressure depletion exceeds 70%, the wave volume decreases significantly, and gas should be injected in time to supplement the formation energy. At present, the study block has been depleted to 72.5%, which has entered the best time for gas injection. The synergistic optimized injection and production parameters are: gas injection rate of 9000 m^3/d, permeability of $0.5\times10^{-3}\mu m^2$, and bottomhole

flow pressure of 9.55 MPa. Under these conditions, the recovery is 39.66%, and the CO_2 sequestration ratio is 27.26%. The research findings provide guidance for the efficient development of CO_2 mixed associated gas drive in tight reservoirs.

Introduction

Carbon Capture, Utilization and Storage (CCUS) technology is a key carbon emission reduction tool, which can realize the dual goals of efficient development of oil and gas resources and synergistic carbon sequestration by capturing CO_2 emitted by industry, then using it for oil and gas extraction[1–4]. Global tight oil reservoirs are rich in resources and have huge development potential[5]. However, due to the complex pore structure and extremely low permeability, conventional development methods are often faced with problems such high initial production, rapid decline, and low utilization of crude oil, and there is an urgent need to explore high-efficiency, low-carbon development paths to enhance the effective utilization of resources[6,7]. In recent years, CO_2 mixed associated gas injection technology has attracted wide attention because of its advantages of both physical and chemical oil driving[8,9]. Compared with pure CO_2 drive, nitrogen drive or methane drive, this technology is more excellent in dissolution and diffusion performance with mixed-phase drive ability, which can more effectively reduce the viscosity and improve the fluidity of the crude oil[10,11]. Meanwhile, the residual oil in the reservoir pore space is utilized through the mechanism of dissolution-extraction, which can effectively enhance the recovery of the reservoir. In addition, CO_2 reacts with the minerals in the reservoir rock, which contributes to the change of reservoir wettability from oleophilic to hydrophilic and further enhances the oil displacement efficiency[12–14]. Combining artificial intelligence algorithms to dynamically optimize the CO_2 mixed associated gas injection can realize rapid screening and adaptive optimization of complex injection parameters[15]. Constructing a scientific injection and production control strategy, accurately identifying favorable driving channels and residual oil distribution, achieving synergistic and integrated optimization of recovery and sequestration ratio. This technology system provides new ideas and solid technical support for the large-scale application of CO_2 mixed associated gas in tight reservoirs[16,17]. For the mixed injection of CO_2 and associated gas, Masalmeh et al. found that the recovery increase of mixed-phase drive can be about 20%, compared with the non-mixed-phase drive, through the continuous injection of 50% methane and 50%, and alternating water-gas injection experiments[18]. Hou et al. conducted the research of N_2-CO_2 mixed-gas cyclic injection in the depletion of thick oil reservoirs, and confirmed that there is favorable synergy between the viscosity reduction effect of CO_2 and the reservoir pressure maintenance ability of N_2, resulting in effective recovery enhancement[19]. Hoffman et al. conducted a comparative investigation of the injection effect of CO_2, non-mixed-phase hydrocarbon gas and mixed-phase hydrocarbon gas in Coulee field, Montana, USA. The findings demonstrated the both miscible hydrocarbon gas and CO_2 drive can significantly improve the recovery of ultra-low permeability reservoirs, in which the miscible of the injected gas a critical factor affecting the drive efficiency[20]. Chen et al. investigated the effect of CO_2 mixed associated gas on the recovery and carbon burial effect by molecular dynamics simulation. The results demonstrated that the recovery and burial ratios were positively correlated with H_2S content, while negatively correlated with the nitrogen content in the gas mixture[21,22]. In addition, Nassabeh et al. utilized deep convolutional neural network for intelligent identification of petroleum reservoir model, with model identification accuracy of up to 98.36% and good noise immunity, which is able to deal with incomplete data and complex conditions of onsite pressure transient signals[23]. Gulbahar et al. systematically combed the application of artificial neural networks in reservoir description, the method was extended to injection and production parameter optimization and other fields, which demonstrated good adaptability and application prospects[24].

The above investigation demonstrated that CO_2 mixed associated gas injection driven by artificial intelligence can not only effectively enhance the recovery and carbon sequestration capacity of tight reservoirs, but also achieve dynamic regulation and optimization of injection parameters. This investigation is the first time that artificial intelligence technology is applied to the CO_2 mixed associated gas replacement process in tight reservoirs, which provides a strong technical support for efficient development and low-carbon transition. The main contents of this investigation include (1) constructing a prediction model for residual oil distribution based on spatiotemporal mixed-attention mechanism. (2) combining artificial neural network and genetic algorithm to optimize the injection parameters and the main control factors. (3) conducting a synergistic optimization study on the oil recovery and carbon sequestration effect of CO_2 mixed associated gas drive.

Reservoir characteristics

Reservoir type

The development of the submarine reservoirs in a field of Bozhong is mainly affected by tectonic movement, weathering and leaching, and rock type, forming two main reservoir types, the weathering zone and the inner curtain zone. The weathering zone is strongly modified by weathering and has high porosity and permeability, while the inner curtain zone is dominated by tectonic cracks and has relatively poor reservoir properties. The locally developed conglomerate section at the top shows obvious thickness and physical property changes. Overall, the physical properties of the reservoir gradually weaken from the weathering zone to the inner curtain zone, which presents significant non-homogeneous characteristics. In particular, fracture is the dominant factor controlling reservoir development, and microtopography plays a moderating role locally. The burial depth of the study block is 1500-3200 m. The results of logging interpretation show that the average total porosity of the weathered zone in the submerged mountains is 6.4%, of which the fracture porosity is 0.65% and the matrix porosity is 5.8%; the average permeability is $5.6 \times 10^{-3} \mu m^2$, and the matrix permeability is $0.35 \times 10^{-3} \mu m^2$.

Reservoir Fluids

The density and viscosity of ground crude oil vary with temperature. The density of crude oil is 0.812 t/m^3 at 20°C, and the density decreases to 0.790 t/m^3 at 50°C, with a viscosity of 0.79 mPa·s. Ground oil has a low viscosity, a high gas-to-oil ratio, and a large volume coefficient (1.85 to 1.89), exhibiting the characteristics of a volatile oil. The composition of the natural gas is dominated by methane in the range of 72 to 81% and intermediate hydrocarbons in the range of 10.87 to 14.18%. The gas contains a high proportion of carbon dioxide (7.49% to 13.05%) and a low content of hydrogen sulfide (18.2 mg/m^3), making it a natural gas with little to low hydrogen sulfide. Ground crude oil has low density and viscosity, which is conducive to improving oil fluidity and production efficiency. Both natural and dissolved gases are rich in methane and intermediate hydrocarbons and low in hydrogen sulfide, which facilitates safe and environmentally friendly gas treatment. The gas-oil ratio is high, the volume coefficient is large, and it has the characteristics of volatile oil, and these characteristics make the formation oil easy to be driven out by gas drive. According to the above reservoir characteristics, this block is suitable for the development of CO_2 mixed associated gas drive. The high percentage of CO_2 in natural gas provides sufficient CO_2 resources for associated gas drive, which contributes to the improvement of oil and gas recovery, and at the same time, reduces CO_2 emission, realizing the double enhancement of economic and environmental benefits.

AI-driven synergistic optimization mechanism for recovery and sequestration

Residual oil prediction based on spatiotemporal hybrid attention mechanism

Methods and principles. In this investigation, an intelligent prediction method for residual oil distribution combining an ono-dimensional convolutional neural network (1D-CNN), a long-short-term memory network (LSTM), with a temporal and spatial mixed-attention mechanism is proposed. Under the guidance of gas-driven oil theory, combined with the systematic analysis of the main controlling factors of residual oil distribution, the sample library for model training was constructed, and data cleaning and fusion work was conducted. In order to improve the expression ability of the model to the complex behavior of the reservoir, it is necessary to systematically classify and extract the input characteristic parameters, so as to ensure that the key driving factors are effectively represented in the prediction. In this investigation, the input parameters are divided into three categories: reservoir physical properties, seepage characteristics and reservoir development parameters. The three types of parameters represent the reservoir static structure, fluid transport characteristics and dynamic development behavior, which have an important role in controlling the distribution of residual oil and need to be considered comprehensively in the model construction. The first category is reservoir physical parameters, including tectonic features, planar and vertical permeability distribution, formation thickness, and rhythmic features. Although these static characteristics change little during the development process, the unevenness of the spatial distribution has a significant impact on the enrichment and distribution of residual oil. The second category is seepage characteristic parameters, which mainly reflect the fluid properties and its transport characteristics in the reservoir, such as the viscosity of the replacement agent, the viscosity of the crude oil, the fluid density, the degree of mineralization, the relative permeability curves of oil and water, and the interfacial tension between oil and water. These parameters can be regulated by engineering measures, which are of direct significance in enhancing oil driving efficiency. The third category is reservoir development parameters, such as instantaneous injection and recovery, cumulative injection and recovery volume, and formation pressure level, which are typical dynamic features that evolve continuously with the development process and have an important influence on the temporal changes of residual oil.

According to a large amount of historical numerical simulation data, 9 typical reservoir models are selected as the basis, and about 180 simulation models are extended by perturbing the distribution of reservoir physical parameters, relative permeability curve, oil-water viscosity ratio, well network configuration and boundary conditions, to construct a high-quality sample library for model training and learning.

In order to screen the degree of contribution of the impact factors, this investigation utilized the Variable Importance method. This method, proposed by Breiman and originally applied to random forest models, is a commonly adopted strategy for importance assessment in machine learning. For a given feature Zi, its variable importance is assessed by the change in the prediction error of the model after randomly replacing the values of the feature. Specifically, setting the dataset [Y, X], where X= [z1, z2, z3……zm], first train the model on the original feature set and calculate its prediction error, using the mean square error (MES) as the loss function. When the prediction is performed again after row-level random replacement of feature zi, if the loss increases significantly, it means that the feature is more important in the model.

$$e_{\mathrm{orig}} = loss(Y, X) = Mean_t\left(\left(y_t - y_{pred,t}\right)^2\right) \tag{3.1}$$

The disturbance prediction error is 3.2.

$$e_{shuf,i} = loss\left(Y, X_{shuf,i}\right) \tag{3.2}$$

Where the feature vector after replacement is denoted as $X_{shuf,I}$ =[z1, z2, z3, …z $_{i}^{shuf}$,….z_m], The i feature Z_i is randomly disrupted, while the rest of the features remain unchanged. By repeating this process of computing the difference between the replacement and prediction several times, the replacement error e_{shuf} can be obtained for the expected value of feature z_i. Finally, the importance (VI) of feature Z_i to target variable Y is defined as 3.3.

$$VI_i = E\left(e_{shuf,i}\right) - e_{orig} \tag{3.3}$$

In this study, Variable importance (VI) was adopted to assess the degree of contribution of each input feature to the prediction target. The input features include variable such as injection rate and formation pressure, and the target variable is oil production. The model prediction error was measured with mean square error (MES). In variable importance analysis, a feature is randomly replaced, keeping the rest of the features unchanged, and predicted using a trained model. Since this substitution breaks the association between the feature and the target variable. If the prediction error increases significantly, it means that the feature has a large impact on the model output. The significance of the variable is measured by the difference between the expected value of the replacement error E (eshuf) and the original prediction error e orig. In order to achieve the contribution comparison between different features, the importance values of all features are normalized. The maximum value is normalized to 1.0, the rest of the features are scaled proportionally to obtain the relative importance level of each feature to the predicted target.

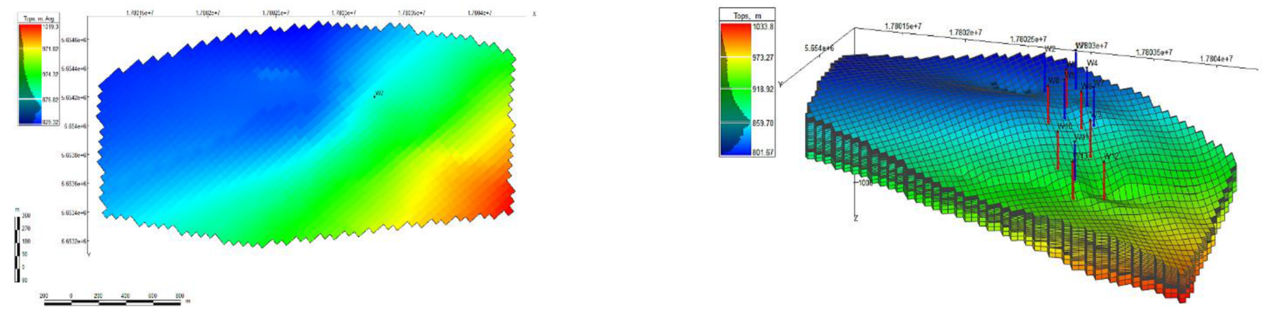
$$new_{VI_i} = \frac{VI_i}{\max\limits_{1<j<N} VI_j} \tag{3.4}$$

VI_i corresponds to the replacement importance of the I gas injection well to the production well, and maxVI_j is the maximum value of the importance of all variables.

Modeling and validation. In this investigation, the tight oil reservoir in a block of Bozhong is taken as the research object, and in order to simulate its development process more accurately and efficiently, t-Navigator, an industry numerical simulator, is adopted to conduct the modeling and simulation. At the same time, the numerical simulation results are compared, with the residual oil distribution predicted by the feature fusion neural network model based on the spatiotemporal hybrid attention mechanism, to verify the feasibility and effectiveness of the neural network in residual oil prediction. The entire modeling process includes the following steps.

Target block data processing. The relevant parameters of the target block were systematically organized and data processed, covering the geological model, well completion data, well history data, as well as petrophysical parameters and fluid PVT data.

Model coarsening. The total area of the numerical simulation investigation is 4.17 km^2. In order to improve the computational efficiency and keep the accurate expression of geological features, the numerical model adopts the grid system after the coarsening of geological modeling results. The surface grid is 61×53, with 3233 cells, and the grid step is 50 m×50 m. The vertical direction is divided into 43 layers to better characterize the non-homogeneity and stratification of the reservoir. The total number of cells in the coarsened 3D grid model is 139019, which effectively improves the simulation computation efficiency on the basis of geological authenticity and numerical stability, and the coarsened model is shown in Fig 1

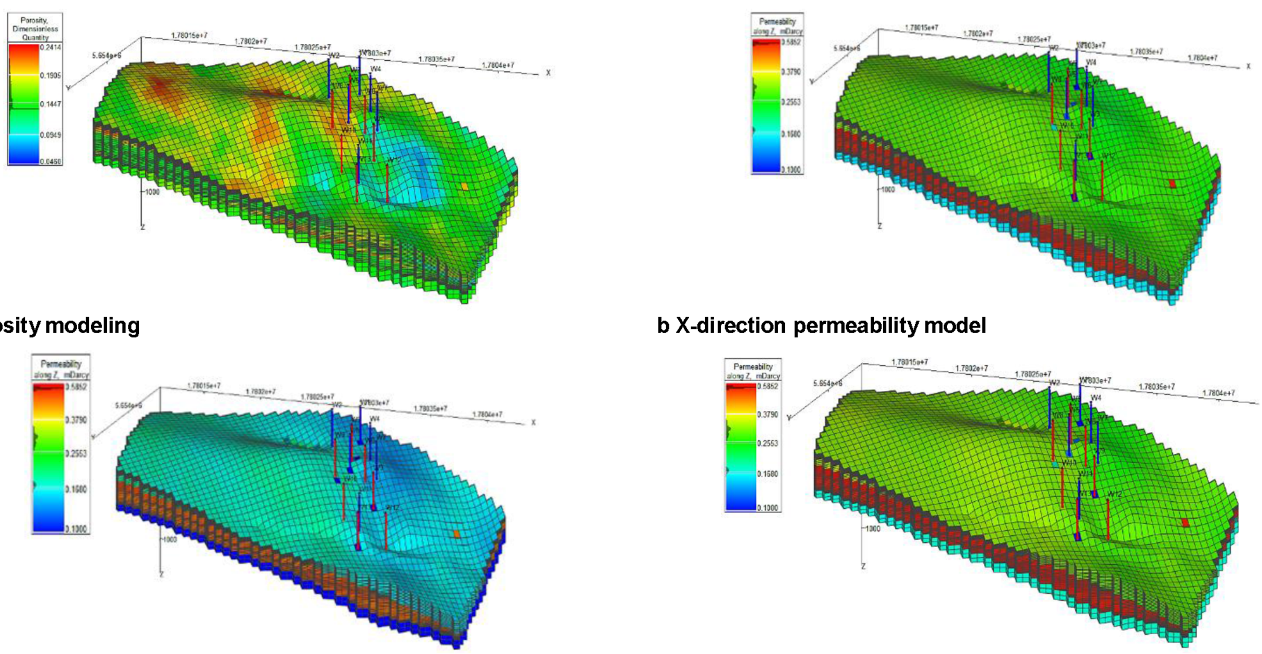


a Reservoir top tectonic map

b Reservoir 3D map

Figure 1—Geological model roughened map

Reservoir parameter coarsening. The grid parameter field is constructed by interpolation method on the basis of the geological model. The physical properties of the reservoir mainly include the spatial distribution of key parameters such as porosity, permeability and oil saturation. The distribution of porosity and permeability is displayed in Fig 2.



a Porosity modeling

b X-direction permeability model

c Y-direction permeability model

d Z-direction permeability model

Figure 2—Coarsening of reservoir physical properties

Production dynamics fitting. The production dynamics fitting is a crucial part of the history fitting work. The core objective is to adjust the reservoir related parameters, making the simulation results fit the actual production dynamics at the current stage. In the fitting process, this research follows the principle of parameter adjustment, to ensure the rationality of the model. The fitting results are shown in Table 1, the fitting error of reservoir is 1.28%, and the fitting error of cumulative oil production is 4.96%. The model fitting results are relatively good.

1. The adjustment of reservoir permeability is controlled with 5 time, ensuring the simulation accuracy and the credibility of the geological model.
2. The porosity and effective thickness are only finely adjusted to a limited extent, avoiding non-physical alterations to the reservoir structure.

3. The well index is adjusted by a factor of no more than 10, reflecting a reasonable difference in wellbore and formation connectivity.
4. Depending on the fit, the phase permeability curve and the capillary pressure curve are optimized locally or holistically.
5. The parameters such as the conductivity coefficient, initial saturation, fluid distribution, and fluid physical properties were slightly adjusted, for further improvement of the fitting accuracy.

Table 1—Parameter fitting results

Fitting comparison	Geologic reserves/10^4t	Cumulative oil production (10^4t)
Actual data	499.45	1.4263
Calculation result	493.04	1.4971
Relative error	1.28%	4.96%

Residual oil distribution prediction. After three stages of development of the target block, the oil saturation around the production wells decreased significantly, forming a relatively stable pattern of residual oil distribution, and its saturation distribution is demonstrated in Fig 3. In term of the overall average distribution, the residual oil after the CO_2 mixed associated gas drive shows obvious non-uniformity characteristics, with the western region having a relatively high residual reserve, and the eastern region having a relatively low residual reserve. Combined with the spatial distribution characteristics of the residual oil, it can be categorized into the following five types:

1. Well network imperfection-type residual oil: Owing to the insufficient arrangement of the well network, some areas have not been effectively utilized;
2. Thin differential layer controlled residual oil: controlled by the distribution of thin differential layers in the reservoir, making it difficult for fluids to effectively enter this part of the region;
3. Sealed fault sheltered residual oil: fault structures are closed, hampering the entry of gas into the fault-enclosed stagnant zone;
4. Insufficient control of well network type residual oil: although well network coverage is available, the injection and production system does not have sufficient control over the area, and the efficiency of the displacement is low;
5. Sporadic sand body control type residual oil: limited by the sporadic distribution of sand bodies, resulting in discontinuous replacement and residual oil remaining in isolated belts.

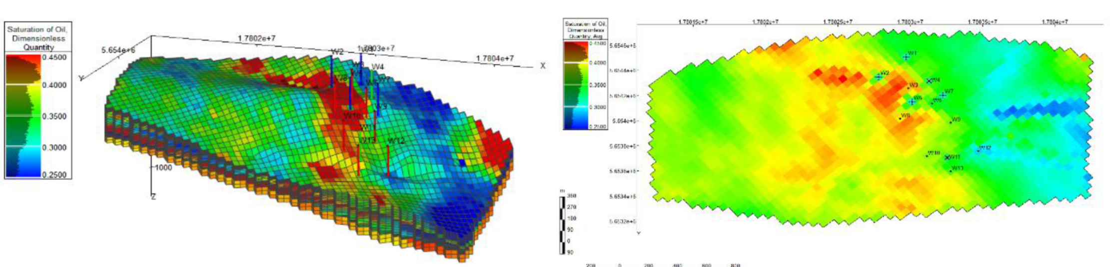
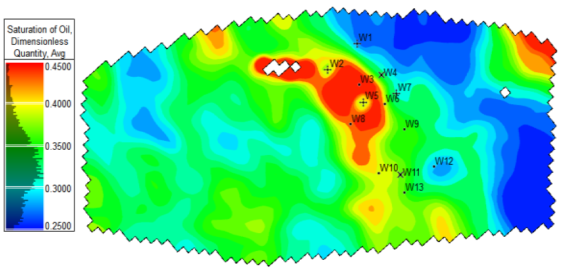
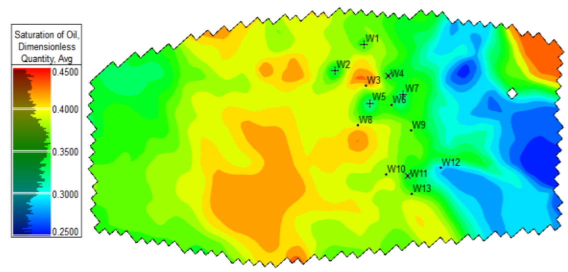


Figure 3—Residual oil saturation distribution

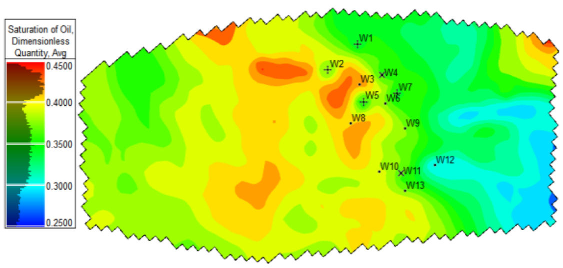
From the distribution of residual oil in each small layer (Fig 4), some layers have shown obvious water-phase breakthrough phenomenon, which is affected by significant interlayer non-homogeneity. The degree of mobilization of different small layers varies greatly, and the overall performance is obviously unbalanced. Taking the old area an example, the B3c, B4a, B4b and B4c layers still maintain high oil saturation due to the poor physical properties of the reservoirs, and the degree of the recovery is low. In contrast, the B3a, B3b and B5 layers are less saturated with oil and have a relatively high degree of utilization. Comprehensive analysis indicates that the B3c, B4a, B4b and B4c layers are important layers for residual oil enrichment, with good development potential, and are the main target layers for CO_2 displacement, while the A2, A3 and A4 layers have small original geological reserves and limited development potential. Therefore, how to further increase the utilization degree of the main layer, and to effectively activate the residual oil in the thin differential layer is the key task and direction of this CO_2 driving development.



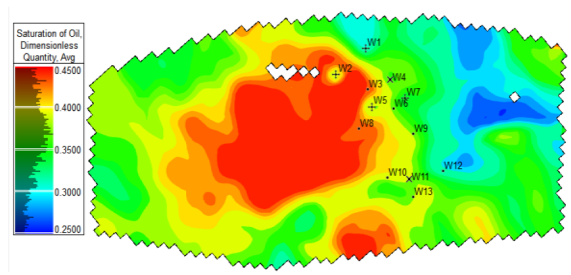
A 4 Residual oil distribution



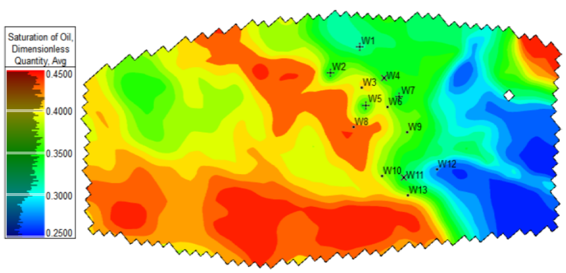
B 3a Residual oil distribution



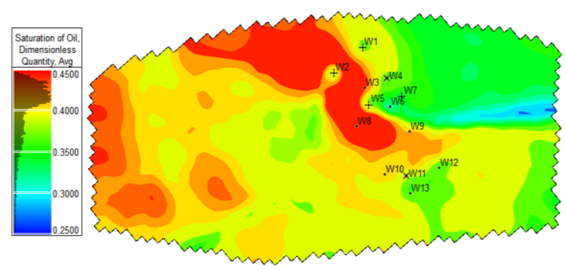
B 3b Residual oil distribution



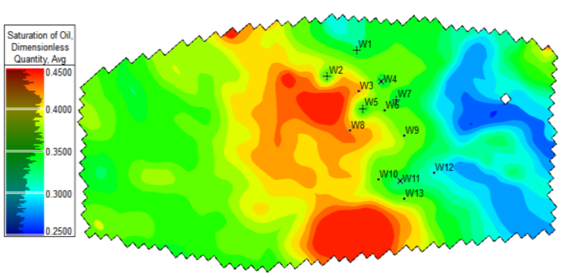
B 3c Residual oil distribution



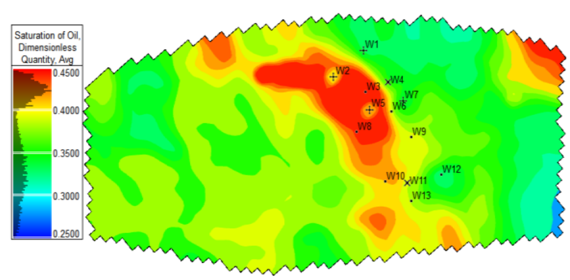
B 4a Residual oil distribution



B 4b Residual oil distribution



B 4c Residual oil distribution



B 5 Residual oil distribution

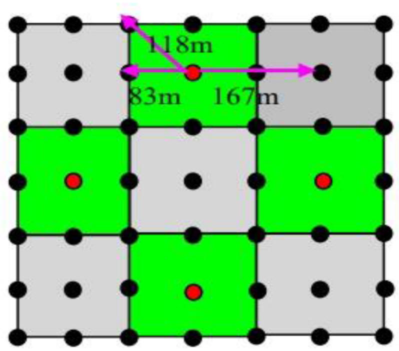
Figure 4—Model coarsening

Parameter optimization of CO_2 mixed associated gas

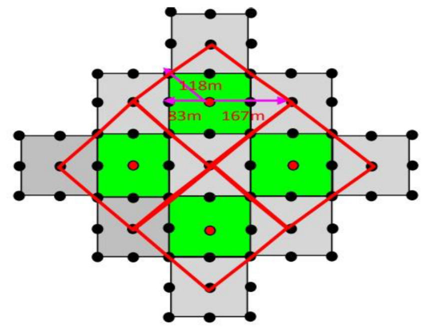
Well network. In the CO_2 drive pilot test, the adopted reverse nine-point well network has problems, such as energy spillover and low CO_2 utilization efficiency. Numerical simulation findings demonstrate that the range of energy spill can reach up to 167 m, indicating that the current injection to extraction well ratio is 3:1, with weak injection energy control capability, which can easily lead to rapid CO_2 breakthrough and insufficient oil displacement. Therefore, it is necessary to appropriately expand the control area of the well group and increase the number ratio of injection and production wells, thus enhancing the control ability of the injected gas. According to the anti-nine-point network model and the analysis of field development conditions, it is believed that the introduction of composite network structure can alleviate the problem of energy spillage, while improving the oil-driving efficiency. Based on this, three optimized well network types are proposed in combination with the study of the anti-nine-point method: anti-nine-point pumping and thinning well network, small -return-shaped well network, and large -return-shaped well network. The distribution of injection and extraction wells in each well network type is shown in Figure 5. At time same time, the parameters of gas injection rate and injection/production ratio were optimized o for each well network form, taking into account the actual conditions of field operation. Through comparing the differences in recovery and economic benefits of different well networks, the well network form with the best performance in terms of improved CO_2 drive recovery and economy was finally selected, which will be the preferred option for the next stage of popularization and application.

At present, the pilot test area adopts the form of reverse nine-point well network, with the arrangement of one gas injection well in the center and eight production wells in the periphery, the control area of the well group is 27,556 m², and the ratio of the number of injection and extraction wells is 1:3. The main features of this well network are the small control area and the low ratio of the injection and production wells, which result in the fast breakthrough of CO_2 and the prone to the phenomenon of energy spillage. The anti-nine-point pumping and thinning network consists of four wells in the inner line. With 8 wells in the periphery, the ratio of injection and production wells is 1:7, and the control area is enlarged to 55,696 m², which is about twice as much as that of the traditional anti-nine-point well network. This well network is more effective in controlling energy spillage, but owing to the small overlap between the inner and outer wells, the local injection/production wells ratio still drops to 1:3, which affects the efficiency of replacement, in case of gas eruptions in the inner wells. The small -return-shaped well network is arranged in the form of 4 wells in the inner line, 12 wells in the periphery, and the number of injection and production wells is 1:9, with a control area of 62,001 m². The advantage of this well network is that after the gas scram in the inner wells, the injection and production ratio (1:5) can still be maintained at a high level, which is convenient for adjusting the reach of CO_2 drive. However, its shortcoming is that the well spacing between the gas injection wells, and the inner line wells is small (about 41.5 m), which is easy to lead to early breakthrough. On the basis of the reverse nine-point well network, a large reverse 17-point well network is added to the periphery, forming an arrangement of 8 wells in the inner line and 16 wells in the outer line. The ratio of injection and production wells is as high as 1:15, and the controlled area of the well group reaches 110,224 m², which has a larger reserve control capacity and higher single-well production capacity. The larger distance between gas injection wells and production wells in this well network is favorable for delaying breakthrough, but due to the need for a larger gas injection rate to maintain driving energy, the bottom of the well is prone to holding pressure; once gas flaring occurs, it is more difficult to seal and repair the gas flaring, which affects the overall effect of oil driving.

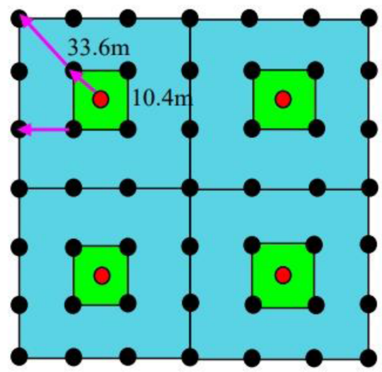
The recovery percentages under different well network forms after 25 years of development are presented in Figure 6. The results indicated that the small gyratory network has the highest recovery performance, while the traditional reverse nine-point network has the lowest recovery, indicating that optimizing the well network structure has a significant effect on improving the CO_2 drive efficiency.



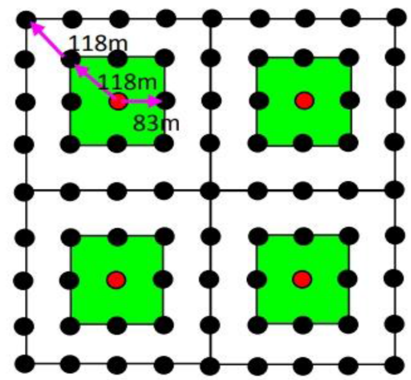
a Reverse nine-point well network



b Anti-nine points pumping thin well network



c Small -return-shaped well network



d Large -return-shaped well network

Figure 5—Schematic diagram of well network types

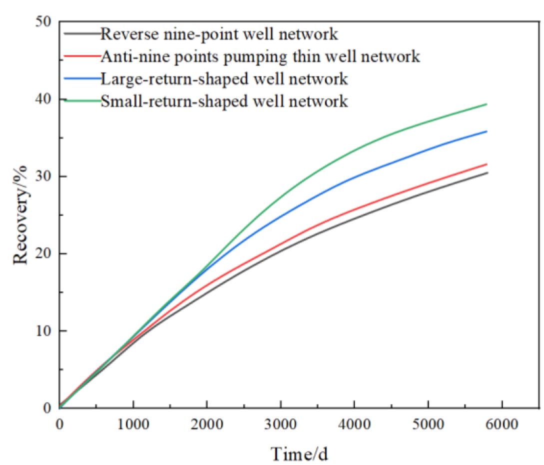
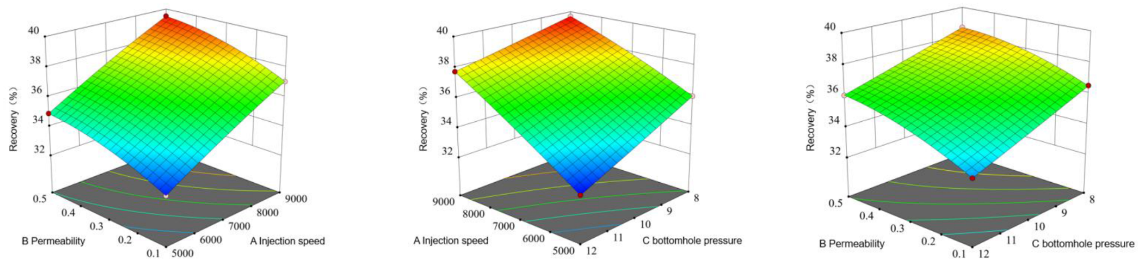


Figure 6—Recovery from different well networks

Controlling factor. With the support of Design-Expert software, a four-factor and three-level response surface test program was designed using Box-Behnken response surface method. Based on the numerical simulation results, the development effect, carbon sequestration capacity, injection and recovery parameter optimization strategy of CO_2-mixed associated gas drive in tight reservoirs are systematically investigated. Through analyzing the sensitivity of recovery and storage ratio to the influencing factors and its interaction, the synergistic optimization of oil drive and storage is conducted. Figure 7 demonstrates the interaction analysis of gas injection rate, reservoir permeability and bottomhole flow pressure on the recovery. The results demonstrate that the interaction between gas injection rate and reservoir permeability is not significant. The interaction between reservoir permeability and bottomhole flow pressure is significant, and the overall effect is moderate. The one-way analysis showed that the recovery was positively correlated with the reservoir permeability, and negatively correlated with the bottomhole flow pressure, with the bottomhole flow pressure having the most sensitive effect on the recovery. The interaction between gas injection rate and bottomhole flow pressure has a significant effect on the recovery. With the increase of gas injection rate, the recovery increases continuously; at the same time, the decrease of bottomhole flow pressure can also promote the recovery increase, but the magnitude of its effect is relatively small. Combining the performance of all factors within the test level, the optimal combination of gas injection parameters is: gas injection rate of 9000 m^3/d, reservoir permeability of 0.5×10^{-3} μm^2, and bottomhole flow pressure of 8 MPa, at which the recovery reaches the highest value.



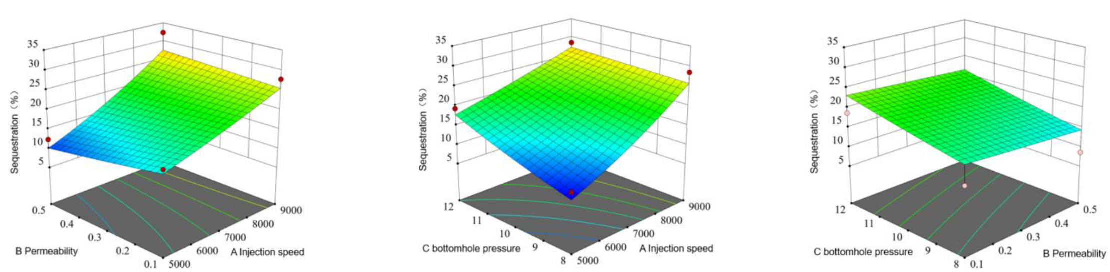
a Gas injection rate and permeability

b Injection rate and bottomhole flow pressure

c Permeability and bottomhole flow pressure

Figure 7—Interaction of gas injection rate, bottomhole flow pressure reservoir permeability and recovery

Figure 8 demonstrates the interaction analysis of gas injection rate, reservoir permeability, and bottomhole flow pressure on burial sequestration ratio. The findings indicated that the interactions between gas injection rate and reservoir permeability, bottomhole flow pressure and reservoir permeability did not have a significant effect on the sequestration ratio, indicating that these two groups of factors have a weak joint modulation effect on the sequestration ratio within the test range. The overall fluctuation of the sequestration ratio is small with the change of gas injection rate and permeability. In contrast, the interaction between gas injection rate and bottomhole flow pressure had a significant effect on the sequestration ratio. The specific performance is as follows: with the decrease of gas injection rate and bottomhole flow pressure, the sequestration ratio increases continuously, and the influence of gas injection rate significantly larger than the bottomhole flow pressure. Comprehensive analysis indicates that, within the range of parameters of this test, the optimal gas injection conditions for realizing higher sequestration rate are: gas injection rate of 9000 m^3/d, bottomhole flow pressure of 12 MPa, and reservoir permeability of 0.5×10^{-3} μm^2.



a Gas injection rate and permeability

b Injection rate and bottomhole flow pressure

c Permeability and bottomhole flow pressure

Figure 8—Interaction of gas injection rate, permeability, bottomhole flow pressure, and sequestration ratio

Injection timing optimization. In the early development concept, CO_2 drive mainly works by expanding the wave volume, increasing the formation pressure. A large number of field test results indicated that the timing of CO_2 injection has a significant impact on the final recovery. The earlier the gas injection is initiated, the more likely it is to realize the development goal of "getting gas early, getting gas fast, and getting more gas", which also contributes to the effective burial of CO_2. However, the timing of gas injection needs to be considered on balance: too late gas injection may limit the ability of CO_2 to drive crude oil, reducing the potential for enhanced recovery; while too early gas injection may not fully utilize the natural formation pressure, affecting the dissolution and burial efficiency of supercritical CO_2. Therefore, scientifically and reasonably determining the timing of CO_2 injection is the crucial to realize oil and gas production increase and carbon sequestration.

In the process of reservoir development, the pore structure and permeability of the reservoir evolve continues to increase, which results in a significant increase in the difficulty of profile regulation at different development stages. With the wide application of CO_2 drive technology, it is widely recognized that the main mechanism is to expand and wave volume, followed by improving the drive efficiency. In the later stage of gas drive development, fluid migration and pore blocking phenomena cause by gas injection will lead to changes in porosity and permeability. The research demonstrated that the permeability of the middle and high permeability layer is enhanced by 10% to 40%; while the low permeability later decreases by 5%to 25%; the non-homogeneity within the layer is significantly enhanced, the difficulty of profile improvement is increased, and the ability of the wave volume is significantly weakened. Accordingly, domestic scholars have statistically determined the relationship between gas injection timing and wave volume (Table 2, Figure 9). The plotted relationship curve shows that when the formation pressure failure exceeds 70%, the wave and volume decrease rapidly, and the CO_2 drive efficiency is weakened substantially. At present, the pressure depletion of this block has reached 72.4%, which is close to the inflection point of wave and volume expansion and decay. Continuing to delay gas injection at this stage will significantly reduce the CO_2 drive potential, and timely initiation of gas injection operation is the key to improving recovery.

Table 2—Timing of gas injection in relation to expanding wave and volume

Injection time	Expanding the volume of the wave
Formation pressure depletion 60%	12.09
Formation pressure depletion 80%	10.56
Formation pressure depletion 85%	9.64
Formation pressure depletion 90%	8.16

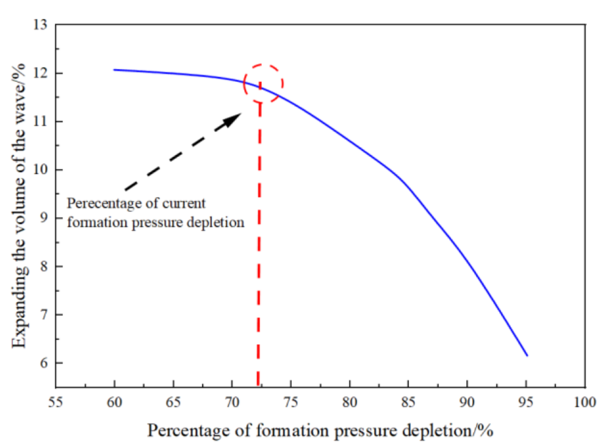


Figure 9—Timing of Gas Injection and Expanded Wave Volume

Synergistic optimization of CO_2 mixed associated gas drive and sequestration integration

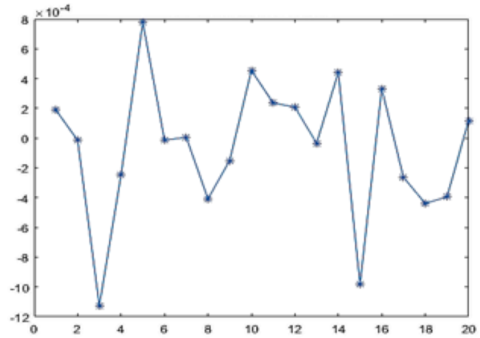
Block sequestration adaptation assessment. CO_2 mixed associated gas has special physical properties, which may cause alteration of the original hydrogeological condition of the reservoir, thereby bringing about the risk of CO_2 burial site, the following critical conditions need to be fulfilled.

1. The temperature and pressure at the sequestration site should meet or exceed the requirements of the CO_2 supercritical fluid state.
2. Space to support large-scale CO_2 storage should be available in the vicinity of the storage area.
3. The permeability of the reservoir region should not be too high, ensuring CO_2 can remain in the subsurface for a sufficiently long detention time.
4. Potential leakage and transport pathways (faults, fractures) should be undeveloped or minimally developed.

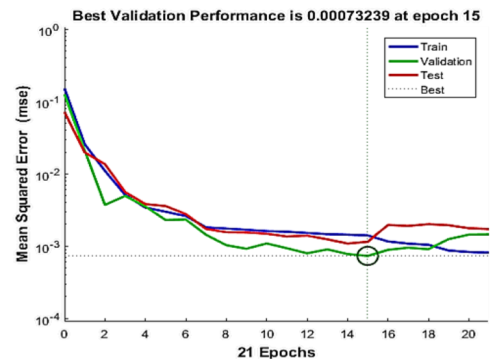
The temperature of the target block ranges from 79.5 to 125.5°C, the pressure ranges from 30.63 to 55.13 MPa, both of which are much higher than the critical pressure (7.38 MPa) and the critical temperature (31.1°C) of CO_2. The average porosity of the zone is 10.6%, and the average permeability is 0.33 $\times 10^{-3}$ μm^2, which ensures sufficiently long retention time of CO_2 in the subsurface. In summary, the study work area basically satisfies the key conditions for CO_2 burial.

Artificial Neural Networks Coupled with Genetic Algorithms to Optimize Parameters. Currently, artificial intelligence technology has become an important frontier direction in the field of reservoir development. On the basis of the above response surface and numerical simulation methods, to achieve the comprehensiveness and objectivity of the gas injection parameter selection process. This investigation constructed a BP neural network model and combined with the genetic logarithm (GA), the output of the neural network gas globally optimized, to obtain the optimal combination of gas injection parameters. In order to ensure the accuracy and robustness of neural network training, sufficient data samples should be provided for learning and training. Based on the numerical simulation results and response surface interpolation data, a total of 200 sets of sample data are selected as inputs, which improves the generalization ability of the model. Considering that the genetic algorithm is applicable to single-objective optimization problems, the optimization objectives need to be normalized. Therefore, in the optimization process, the two indexes of "enhanced recovery" and "storage ratio" are normalized to achieve the synergy and unity between multiple objectives, and enhance the optimization efficiency and practical value. The reservoir recovery response value target value R1 and sequestration ratio corresponding target value R2 are set to obtain the synergistic optimization scheme.

The established BP neural network model has 3 input parameters (reservoir permeability, bottomhole flow pressure, injection rate) and 2 output parameters (recovery, sequestration ratio). The network structure adopts a 3-10-1 architecture, which means 3 neurons in the put layer, 10 neurons in the hidden layer, and 1neuron in the output layer. To ensure that the model has good generalization ability and fitting performance, 180 sets of sample data are selected as the training set, and the remaining 20stes are adopted as the test foe validation. Figure 10 demonstrates the percentage of error in the training process of the neural network and the performance validation results, which indicate that the constructed model can be effectively applied to optimize the has injection parameters. After the completion of neural network training, genetic algorithm (GA) was further introduced to jointly optimize the two comprehensive evaluation indexes of recovery and sequestration ratio. Through the iterative calculation of the genetic algorithm, the convergence curve of the individual fitness with the number of iterations in the optimization process is obtained, the results indicate that the model converges well and the final output is stable. Comprehensive optimization results show that the best combination of gas injection parameters is: gas injection rate of 9000 m^3/d, bottomhole flow pressure of 9.55 MPa, reservoir permeability of 0.5 $\times 10^{-3}$ μm^2. Under this parameter condition, the recovery reaches 39.66%, and the sequestration ratio is 27.26%, which realizes synergistic optimization of gas-driven development, oil displacement and sequestration.



a Percentage of neural network prediction error



b Neural network performance validation

Figure 10—Neural network model prediction and validation

CONCLUSION

In this investigation, artificial intelligence technology combining artificial neural network and genetic algorithm are applied to the optimization of injection CO_2 mixed associated gas drive parameters, which provides technical support for the efficient development of tight reservoirs. The investigation includes (I) Predicting residual oil based on spatiotemporal mixed-attention mechanism. (II) Clarifying the main controlling factors of recovery and storage rate using response surface method. (III) The artificial neural network and genetic algorithm are adopted, to conduct the synergistic optimization research of oil injection and drive and sequestration.

1. The distribution of residual oil in the target block is obvious, and the overall pattern is high in the west and low in the east. The remaining oil is mainly affected by the integrity of well network, thin differential layer, fault closure, injection and recovery control ability and sand body spreading. According to the type, it can be summarized into five types: well network imperfection type, thin differential layer control type, fault blockage type, insufficient control type and isolated sand body control type.
2. The interaction of gas injection rate and bottoming flow pressure has a highly significant effect on the recovery. The interaction between gas injection rate and reservoir permeability was not significant. The interactions between gas injection rate and reservoir permeability, the wellbore flow pressure and reservoir permeability do not have significant effects on the sequestration rate. The effect of gas injection rate and bottomhole flow pressure on sequestration ratio is significant.
3. Three types of well networks are proposed based on the inverse nine-point method well network: the inverse nine-point pumping and thinning well network, the large -return well network, and the small-return well network. The development effect of the small -return zigzag well network at the end of the forecast period is finalized.
4. The recovery and sequestration ratio were optimized based on the synergistic optimization of artificial neural network and genetic algorithm, the optimal parameters were determined as: gas injection rate of 9000 m^3/d, permeability of $0.5 \times 10^{-3} \mu m^2$, and bottomhole flow pressure of 9.55 MPa.

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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